

Kamran Heydarov

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SUMMARY

Chemoinformatics M.Sc. student working at the interface of physical chemistry, molecular modelling and machine learning. Experienced across the full QSPR workflow: curating experimental data from chemical databases, computing and selecting molecular descriptors, training and validating ensemble models, and deploying them as documented tools that non-specialists can use. Author and maintainer of open-source scientific Python libraries for ionic liquid thermodynamics and thermophysical data access (pyionics, ILThermoGUI, ILQSPR), working in Python with RDKit, Mordred, scikit-learn and XGBoost. This is complemented by quantum chemistry coursework and DFT calculations, and by laboratory and teaching experience in chemical analysis. Particularly interested in applying data-driven methods to the rational design of ionic liquids and sustainable solvents.

EDUCATION

University of Strasbourg

M.Sc. in Chemoinformatics

Baku, Azerbaijan

Expected Jun 2027

Azerbaijan State Oil and Industry University

B.Sc. in Chemical Engineering, Chemical Technologies Faculty

Baku, Azerbaijan

Jun 2025

WORK EXPERIENCE

Azerbaijan State Oil and Industry University

Django Developer & Chief Specialist

Baku, Azerbaijan

Sep 2025 – Jun 2026

- Engineered Django applications and managed database schemas (e.g., MongoDB) to handle critical university data.
- Built and maintain the university's scholarly publishing platforms, front end and back end: press.asoiu.edu.az and the OJS journal system at ojs.asoiu.edu.az.
- Coordinated the migration of *Baku Mathematical Journal* to Elsevier Digital Commons across nine international working meetings with the Elsevier team, covering site structure, metadata preparation and article migration; serve as Executive Editor of the journal.
- Registered DOIs and deposited metadata through Crossref, aligning the university's journals with international indexing requirements.
- Designed and programmed the official website of the MacroFrontiers 2026 international conference (asoiu.edu.az/mpmc2026), and built the electronic submission, tracking and plagiarism-screening system for the NASCO international conference.
- Administered Elsevier research platforms (Scopus, ScienceDirect, SciVal, Reaxys) and Turnitin for ASOIU and UFAZ, and delivered SciVal and Reaxys training seminars to academic staff.

Azerbaijan State Oil and Industry University

Data-Oriented Lab Assistant, supervised by Dr. Leyla Vazirova

Baku, Azerbaijan

Sep 2023 – Feb 2025

- Assisted professors in undergraduate laboratories at UFAZ and ASOIU.
- Guided 20+ students per lab session, improving their understanding of experimental concepts and helping boost lab grades.
- Taught students statistical methods for chemical analysis, enhancing their data interpretation skills and analytical reasoning.

PROJECT EXPERIENCE

ILQSPR: QSPR Modelling of IL-Solvent Mixtures ([code](#), [app](#), [docs](#))

Baku, Azerbaijan

Data Scientist, joint work with Dr. Shamkhal Baybekov ([email](#))

2025 – Present

- Built QSPR models predicting the density and refractive index of pure ionic liquids and their binary mixtures with water, ethanol, and isopropanol as a function of temperature and IL mole fraction.
- Curated training data from the NIST ILThermo database using pyionics, applying duplicate removal, consistency checks, and structure standardisation.
- Represented ionic liquids with Mordred 2D descriptors computed separately for cation and anion, followed by variance- and correlation-based feature selection.
- Trained consensus ensembles of five XGBoost models using group-based 5-fold cross-validation on IL SMILES, reaching $R^2 = 0.90\text{--}0.96$ across all eight property-solvent systems; manuscript in preparation.
- Deployed the models as a public Streamlit application with online documentation, so that non-specialists can obtain predictions directly from SMILES, temperature and composition.

pyionics Python Library ([link](#))

Owner & Developer

Baku, Azerbaijan

Mar 2025 – Present

- Developed a Python-based tool for thermodynamic property estimation of ionic liquids and mixtures.
- Integrated models for phase equilibrium and heat capacity prediction to support research and education.
- Optimised the codebase for usability in academic and industrial thermodynamic analysis.

marimo4obs, Obsidian Plugin (code)	Baku, Azerbaijan
<i>Owner & Developer</i>	2026
- Built an open-source Obsidian plugin that registers the .py extension and opens marimo reactive Python notebooks inside the vault, each backed by a local marimo server. - Supports both the full reactive editor and app mode, and installs marimo automatically when missing. - Distributed as a community plugin with packaged releases and documented installation.	
Chron Shift, Browser Game (play)	Baku, Azerbaijan
<i>Creator & Developer</i>	2026
Developed a single- and multiplayer browser platform game with running and fighting mechanics; selected with it as one of the top 15 studios for the Xsolla-IDDA Baku game incubator.	
ChemGraph, Process Simulation Software	Baku, Azerbaijan
<i>Executive Director</i>	Feb 2025 – 2026
- Led the development of ChemGraph, an open-source PFD design simulator supported by Azerbaijan State Oil and Industry University. - Oversaw its application in education and research, aiding students and researchers in process simulation. - Enhanced the platform's functionality to improve accessibility and usability for academic and industrial applications.	
ILThermoGUI	Baku, Azerbaijan
<i>Data Scientist, group of Dr. Shamkhal Baybekov (email)</i>	Sep 2024 – 2026
- Developed a Python library for installing and accessing NIST ILThermo data. - Cleaned, processed, and structured datasets for improved usability. - Designed visualisations to analyse and interpret thermophysical data, and conducted statistical analysis to extract meaningful insights. - Researched ionic liquids, focusing on their thermophysical properties and applications.	
MentorCorp	Baku, Azerbaijan
<i>Co-Founder & CTO</i>	Jan 2024 – 2026
- Developed a user-friendly interface for the mentoring platform and streamlined resource access. - Achieved an average onboarding time reduction of 30% for new users seeking guidance in professional growth.	
YOUNGH	Baku, Azerbaijan
<i>Founder & CEO</i>	Jan 2024 – Sep 2024
- Built a Q&A platform with 200+ active users, enabling students to earn up to \$500 collectively in its first six months. - Enabled event tracking and provided opportunities through the website. - Sold the platform to a company in our competition, expanding its reach and impact.	

LEADERSHIP & OUTREACH

Science Club	Baku, Azerbaijan
<i>Founder</i>	2023 – 2026
- Founded and ran a student science club organising academic competitions and scientific outreach events. - Organised Integration Bee 2023 Azerbaijan, an integral-calculus competition for university students. - Created AnimAI, a prompt-engineering and generative-image competition, and co-organised GenerO with the UFAZ Chemical Engineers Club, a challenge on the synthesis of oxygen in the Martian atmosphere.	

INCUBATION PROGRAMS

SABAH.HUB Incubation Project Owner, YOUNGH	Jan 2024 – May 2024
UFAZ IGNITE Incubation Program Project Owner, YOUNGH	Oct 2023 – Jul 2024

SKILLS & LANGUAGES

Cheminformatics & ML RDKit, Mordred descriptors, scikit-learn, XGBoost, QSPR modelling, KNIME
Computational chemistry Density Functional Theory for molecular quantum dynamics, thermodynamic property estimation
Programming Python (pandas, NumPy, matplotlib, tkinter), Django, Streamlit, JavaScript, MongoDB, Git, $\LaTeX$
Other tools MATLAB, SolidWorks, Excel
Languages Azerbaijani (native), Turkish (native), English (fluent), Russian (B1)

CERTIFICATIONS

Data & programming Data Manipulation with pandas, MongoDB Python Developer Path, Intermediate Python, JavaScript (Intermediate), GitHub Foundations, KNIME Fundamentals, Excel Fundamentals
Process & laboratory safety ELA982 SACHE® Managing Lab Process Safety 1, ELA986 SACHE® Managing Lab Process Safety 2, ELA954 SACHE® Certificate Program: Lab Safety